

An Internship Report
on
**MANAGEMENT STRATEGIES FOR ENHANCING
FLOOD RESILIENCE IN BIHAR**



Submitted to
**NATIONAL INSTITUTE OF DISASTER MANAGEMENT
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Chapter 1: Introduction

1.1 Background of Flood Risk in Bihar

Bihar's geography places it at high flood risk because it lies downstream of major Himalayan river systems. A number of notable river systems such as Kosi, Gandak, Bagmati, and Mahananda cross state boundaries and are born from faraway Tibetan and Nepalese highlands. Bihar is among India's most flood-prone states, particularly in terms of the proportion of its geographical area exposed to flooding. Bihar's geography places it at high flood risk because it lies downstream of major Himalayan river systems. According to the Bihar State Disaster Management Authority's District Disaster Management framework, approximately 73.06 per cent of the state's geographical area (68,800 km² out of a total 94,160 km²) is vulnerable to flooding ([BSDMA, 2026](#)).

The state's own Disaster Risk Reduction Roadmap (2015-2030) records that 28 of Bihar's 38 districts are flood-prone, and that the state accounts for 17 per cent of India's flood-prone area and 22 per cent of the country's flood-affected population ([Government of Bihar, 2015](#)).

Bihar government reports ([BSDMA, 2026](#)) that river discharge may increase substantially during the monsoon. In combination with low gradients, sediment deposition and channel instability, these conditions may increase the likelihood of overbank flooding, embankment stress and, in some locations, channel shifting.

During the Incessant Ganga seminar held in Patna on 25-26 February 2017, experts called for an immediate review and comprehensive assessment of the Farakka Barrage to restore the Ganga River. They highlighted concerns over reduced river flow, increased siltation, and recurrent flooding associated with the barrage. Similar concerns were raised by the then Chief Minister, who attributed declining downstream flow and increased sediment deposition in the Ganga to the Farakka Barrage ([SANDRP, 2019](#)).

Table 1. Districts with Highest Recurrence of Severe Floods (2001-2022) ([Kumar, 2025](#))

District	Flood Recurrence (Years Affected)	Total Estimated Affected Population (millions)	Major Rivers Involved
Madhubani	17	10.2	Kamla-Balan
Darbhanga	16	9.8	Bagmati
Muzaffarpur	15	9.4	Burhi Gandak
East Champaran	15	8.7	Gandak
Sitamarhi	14	7.6	Lakhandei
Samastipur	14	7.1	Burhi Gandak
Saharsa	13	6.9	Kosi
Gopalganj	13	6.5	Gandak

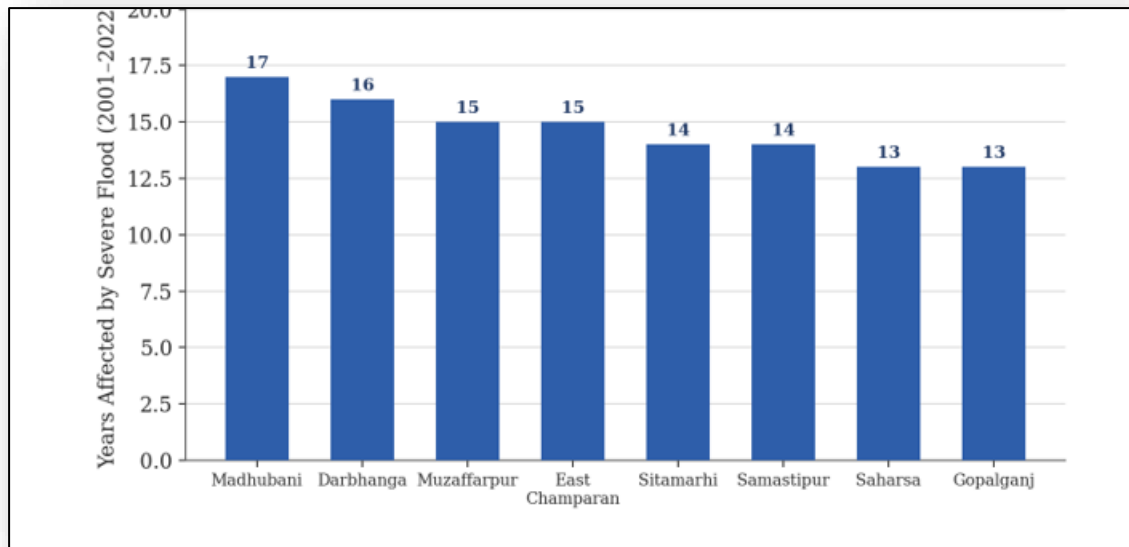


Figure 1. Flood Recurrence by District, 2001-2022. (Author's visualisation of Table 1 data. Adapted from Kumar (2025).

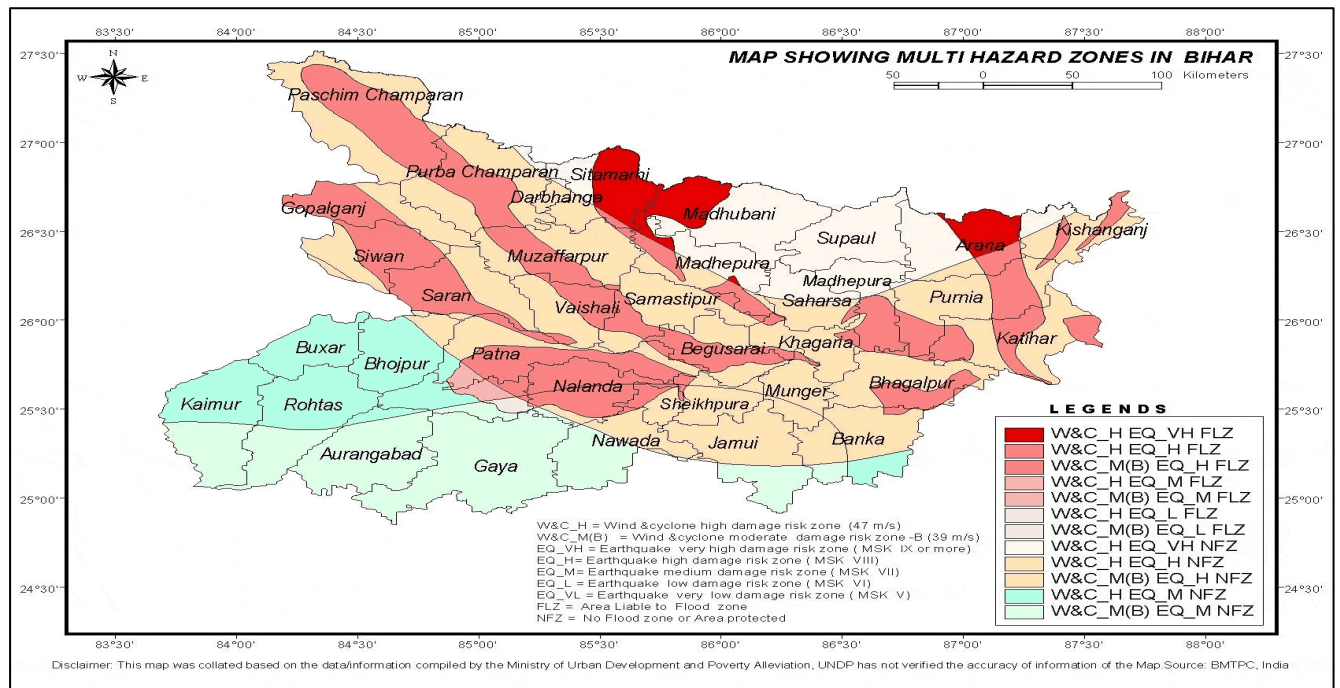


Figure 2. Map Showing Catchment of Major Rivers in North Bihar ([Kansal et al. al 2017](#))

1.2 Rationale of the Study

Bihar's northern districts, situated in the Kosi, Gandak, Bagmati, and Ganga river catchments, are among the most flood-prone regions in India, with the Kosi basin being especially vulnerable because its channels and adjoining floodplains lie at nearly the same elevation, so the river's low gradient combined with heavy sediment infilling causes flooding even under normal rainfall conditions. [Sahani et al., 2023](#)) report that flood frequency, magnitude and duration in the Kosi catchment have increased despite a long-term decline in overall rainfall. The authors associate this

pattern with factors including embankment effects, sediment dynamics and the differing exposure of communities inside and outside embanked areas.

The literature reviewed for this report identifies continuing gaps in preparedness resources, local capacity and embankment/protective infrastructure in some flood-prone communities ([Kumar et al., 2026](#)).

1.3 Objectives

The study aims to: -

1. Evaluate Bihar's flood disaster management framework.
2. Assess flood preparedness, response, and relief mechanisms.
3. Identify challenges and recommend strategies for flood resilience.

1.4 Methodology Adopted

This study adopted a descriptive qualitative research design based on secondary sources, including official reports, policy documents, institutional publications and relevant research literature. A systematic literature review and document analysis were undertaken, and the collected information was analysed thematically in accordance with the study objectives and chapter structure. The study aimed to identify key challenges in Bihar's flood disaster management system and formulate evidence-based recommendations for improving flood preparedness, response and resilience.

Chapter 2: Understanding Floods in Bihar

2.1 Flood Vulnerability Profile of Bihar

Bihar is one of the most flood-prone states in India due to its low-lying alluvial plains, dense river network, and heavy southwest monsoon rainfall. After the bifurcation of the state, Bihar became the most flood-prone state in the country in terms of the proportion of land susceptible to flooding. Approximately **68,800 km² (73.06%)** of the state's geographical area is flood-prone, accounting for nearly **17.2% of India's total flood-prone area** ([NIDM, 2008](#)).

Flood vulnerability is particularly severe in **North Bihar**, where major transboundary Himalayan rivers—including the **Kosi, Gandak, Bagmati, Kamla-Balan, Burhi Gandak, Mahananda, and the Adhwara river system**—enter the state from Nepal. These rivers carry large volumes of water and sediment during the monsoon season. The sudden reduction in river gradient upon entering the plains causes heavy sediment deposition, river course instability, embankment breaches, and prolonged inundation, resulting in frequent and severe flood events ([NIDM, 2008](#)).

Although Bihar has strengthened its early warning systems, flood forecasting, GIS-based flood mapping, and institutional mechanisms, the sources reviewed identify continuing challenges in coordination, data integration, relief logistics, infrastructure resilience and technology. Strengthening integrated flood risk management, real-time information sharing, community-based preparedness, and inter-agency coordination is essential for improving Bihar's long-term flood resilience ([BSDMA, 2016](#); [PIB Delhi, 2026](#)).

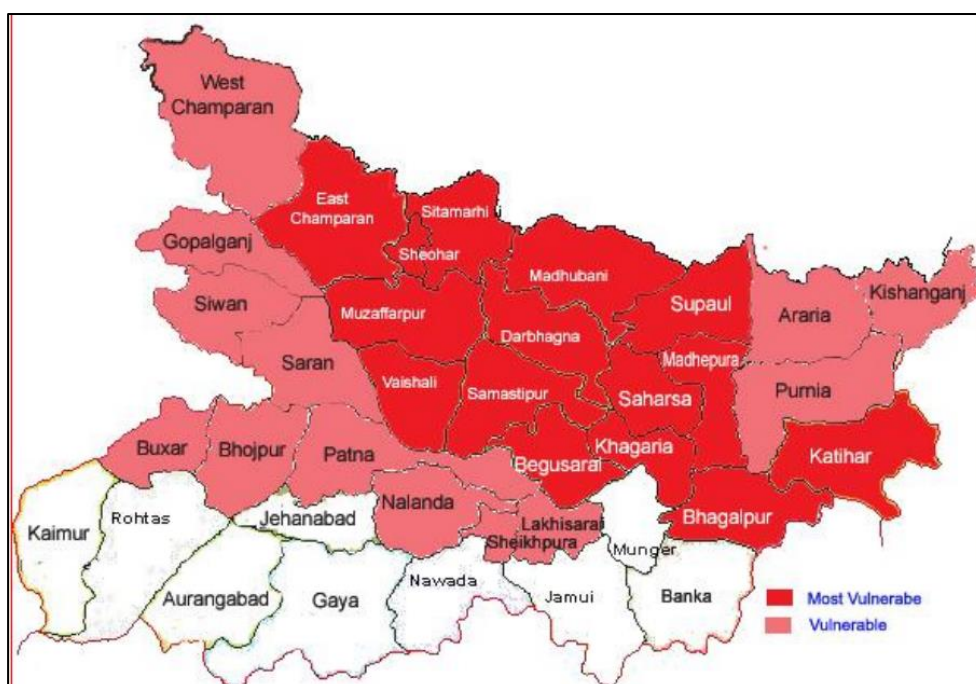


Figure 3. Flood-prone Areas in the state of Bihar ([BSDMA, 2016](#))

2.2 Major River Systems and Flood Dynamics

Bihar is drained by the **Ganga River**, which flows from west to east across the state and serves as the principal drainage system. The Ganga receives several major Himalayan tributaries, including the **Kosi, Gandak, Burhi Gandak, Bagmati, Kamla-Balan, Ghaghara (Saryu), and Mahananda**, as well as southern tributaries such as the **Son, Punpun, Kiul, Harohar, Badua, Chandan, and Karmanasa** ([NIDM, 2007](#)). The Himalayan rivers contribute the majority of

floodwater because they originate in the Nepal Himalayas, where intense monsoon rainfall generates high river discharge and sediment loads ([CWC, 2019](#))

Among these rivers, the **Kosi River**, popularly known as the "*Sorrow of Bihar*," is the most dynamic due to its exceptionally high sediment load, frequent channel migration, and recurring embankment breaches. Likewise, the **Gandak, Bagmati, Kamla-Balan, Burhi Gandak, and Mahananda** rivers normally become inundated during the southwest monsoon, causing massive floods in northern Bihar and affecting districts such as Supaul, Saharsa, Madhepura, Madhubani, Darbhanga, Sitamarhi, Muzaffarpur, Khagaria, Araria, Kishanganj, Katihar, and Purnia ([NIDM, 2007](#)).

Among all these systems, the Kosi has the highest impact on flood risk. Its area of drainage is roughly **74,030 km²**, but about **11,410 km²** of this region belongs to India, while the remaining areas are located in Tibet and Nepal. The Kosi is a perennial river as it receives its water from the melting snow in the Himalayas. Its extremely high silt load has made the Kosi change course by several hundred km in the last 250 years.

Bihar **flood dynamics** rely not so much on river flow alone but rather on the combined impacts of rainfall concentration, channel instability, backwater pressure from the Ganges, drainage blockage and levee collapse. During the monsoon season, rivers originating in the Himalayas and foothills can experience a dramatic rise in discharge, and their flows often increase many times over the non-monsoon level, producing widespread overbank flooding. Flooding in Bihar is therefore not a single-process event; it commonly results from synchronized rainfall in upstream catchments, local precipitation in the plains, and poor drainage escape due to the low slope of the land and obstruction created by roads, railways, canals, and embankments ([Sinha et al. 2012](#)).

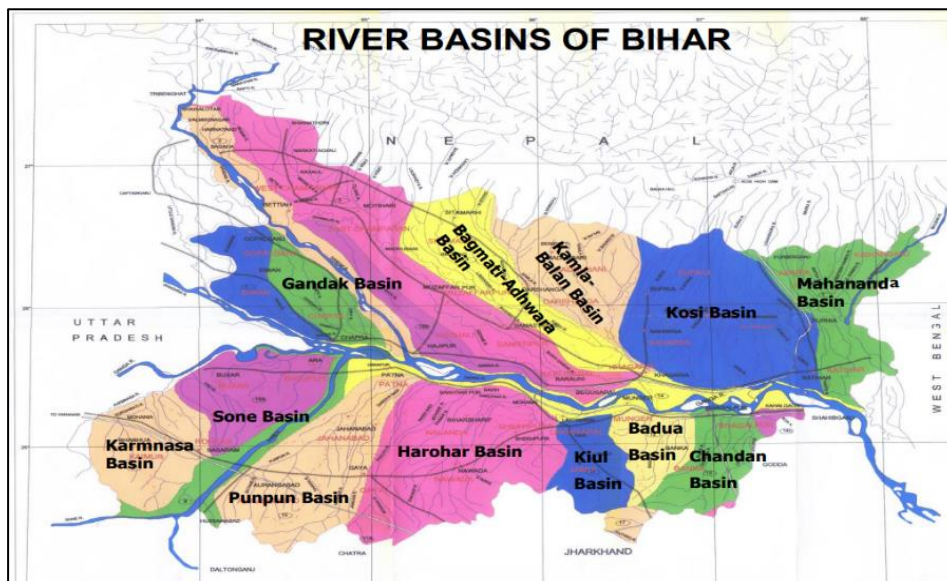


Figure 4. Major River Basins of Bihar ([Vyas J., 2021](#))

2.3 Historical Flood Events in Bihar

Bihar has experienced recurrent and severe flooding for decades, with major flood years including 1978, 1987, 1998, 2004 and 2007 Bihar witnessed high magnitudes of flood ([FMISC, 2025](#)). Research done by Kale (1997) shows that north Bihar plains have experienced the maximum number of floods in the past three decades. ([FMISC, 2025](#)).

1954 flood: - One of the earliest major post-independence flood events was the 1954 flood, which affected North Bihar extensively. According to a field report, it inundated about 2.46 million hectares, affected 7.61 million people, damaged crops over 15.96 lakh hectares, destroyed 1,79,451 houses, and caused 63 deaths ([NIDM, 2007](#)).

1974 flood: - more widespread, affecting both northern and southern districts, including Munger and Santhal Parganas, with severe crop loss and destruction of more than 5 lakh houses ([NIDM, 2007](#))

1997 flood: - The year 1997 saw the occurrence of floods in India as torrential rains caused excessive flooding in Bihar. Heavy rainfall in Madhubani in July caused the destruction of paddy crop as well as the electrical infrastructure; the flash floods in Bagmati River destroyed an area of 6000 hectares of crops, and caused the death of 15 people in Katihar and Muzaffarpur. Due to wall collapse from heavy rainfall, five people lost their lives in Patna on July 21, while the floods' peak was in August, causing the death of 65 people in 20 districts of the state. Over the entire year, the flooding in Bihar impacted 6.965 million individuals, with 1.471 million of land-covered water and 0.655 million hectares ([IMD, 2022](#)).

2008 flood: - There is 2018 Nepal Kosi flood damage of the Eastern Kosi river afflux embankment which resulted in flooding of 5 districts in North Bihar affecting around 30 lakh people and killing over 3000 people. The more recent case of such devastating economic effect was the Kosi flood of 2008 when around 500 villages were flooded in the districts of Supaul, Madhepura, Purnea, Saharsa and Araria. As per Bihar Government estimation, a total of 236,632 houses were completely or partially demolished in the districts of Supaul, Madhepura, Saharsa, Araria and Purnea due to the floods. ([Sinha et al., 2012](#)).

More recent floods show that the hazard remains persistent. The Flood Hazard Atlas of Bihar reports high-magnitude flood years in **2012, 2013, 2016, 2017, 2018, and 2019**, with 2019 standing out because Bihar experienced three flood waves over four months. The occurrence of repeated high-magnitude flood years indicates that significant flood hazard persists despite structural interventions. NRSC identifies upstream rainfall variability, sedimentation and drainage constraints as factors associated with continuing flood impacts ([NRSC, 2021](#))

2.4 Socio-economic Impacts of Flooding

Flooding in Bihar produces severe and recurring socio-economic disruption, particularly in the flood-prone districts of North Bihar where floods erode household assets, reduce income security, disrupt access to services, and deepen pre-existing poverty among vulnerable groups, especially small and marginal farmers, landless labourers, and households below the poverty line. ([GFDRR, 2025](#)).

Floods also have a strong negative effect on **employment and labour markets**. Casual labourers and agricultural workers typically lose access to work during inundation, and income losses can persist for months after the flood because farm operations, transport networks, and local markets remain disrupted. In the post-Kosi context, 93% of surveyed households reported income losses of more than 50% in the first three months after flooding, reflecting the scale of livelihood interruption in flood-affected areas. Similarly seen, loss of production uses like instruments, cattle, and stored goods makes recovery difficult and causes people to borrow money or leave their homes ([UNDP, 2008](#)).

Furthermore, floods cause huge losses of **Housing and assets**. Evaluations at the household level in the state of North Bihar show that the largest part of flood damages is land losses and housing losses, while the second-largest part is loss of household goods, crop production, animal raising, and water and sanitation services ([Megh Pyne Abhiyan & Tata Trusts, 2024](#)). According to an evaluation conducted in 2024, the overall estimated loss registered was Rs 126.3 crore across 2,290 households situated in seven flood-prone regions, with most losses incurred for land and housing costs ([Megh Pyne Abhiyan & Tata Trusts, 2024](#)).

Another important impact is **food insecurity and nutritional stress**. Flood-affected families tend to borrow food from neighbors or relatives, cut back on food consumption, and either sell or

mortgage productive property in the wake of the disaster. The assessment carried out in 2024 shows that over 90% of households affected by floods had to reduce food consumption, with 75% of the families needing to borrow food ([Megh Pyne Abhiyan & Tata Trusts, 2024](#)). Due to the reliance of the majority of rural farms on daily labour and subsistence farming, even a slight disruption in the income stream can lead to dire consequences for nutrition and welfare.

Floods increase **displacements, debts, and relocations**. During and following floods, temporary relocation is usual, with families often settling in shelters, with friends and relatives, or in similar informal settings. According to the 2024 assessment, 82.18% of families faced temporary displacement, while 35.07% pledged their jewellery and almost 25% sold or pledged their livestock to cope with the crisis ([Megh Pyne Abhiyan & Tata Trusts, 2024](#)).

From a broader perspective of development, ongoing flooding works against the growth of human capital by disrupting schooling, health access, and community services. When roads, bridges, drinking water systems, and irrigation networks are damaged, the entire rural economy is affected, and recovery becomes slower and more expensive. Repeated losses and incomplete recovery may contribute to persistent poverty and constrained economic mobility in floodplain districts; however, the extent of this relationship would require longitudinal household and district-level evidence. ([World Bank, 2011](#)).

Overall, the socio-economic impacts of flooding in Bihar are multi-faceted and cumulative. Floods not only cause destruction to crops, houses, and infrastructure but also undermine household resilience through loss of income, debt, forced displacement, food shortages and loss of assets. Thus, flooding in Bihar should be viewed not only as a seasonal calamity but also an underlying development issue causing continuous reproduction of vulnerability and poverty in flood-affected areas ([Megh Pyne Abhiyan & Tata Trusts, 2024](#)).

Table 2. Flood losses in Bihar from 2000 to 2015. ([Kansal, 2017](#))

Year	River which caused flood	Number of Affected					Affected Area		House Damaged		Loss	
		Dist.	Block	Vill.	Human	Animal	Agric	Non-Agric	Fully	Partly	Live	Live-stock
		(Nos.)			(in Lac)		(Million ha.)		(Nos.)		(Nos.)	
2015	Gandak, Kosi	8	32	237	4.850	0.140	0.0090	0.0030	157	12	27	1
2014	Gandak	20	116	2018	30.000	1.000	3.6000	0.3000	3235	2977	158	34
2013	Gandak, Kosi	20	126	4810	72.340	5.600	1.6800	0.6900	9344	117572	253	6480
2012	Gandak, Ganga	11	28	277	2.400	0.080	0.0968	0.0100	322	151	15	0
2011	Ganga	25	166	3577	71.430	6.627	3.5420	0.2877	25840	59342	249	183
2010	Kosi	9	44	679	7.180	3.570	0.1602	0.0390	6187	8983	32	0
2009	Bagmati, Gandak	16	91	1546	22.030	1.346	0.1710	0.9339	3867	3807	97	2
2008	Kosi	18	116	2585	49.952	12.166	0.6405	0.2119	104965	151461	258	878
2007	Gandak, Kosi	22	269	18832	244.420	27.130	1.3323	0.5510	369360	414968	1287	2423
2006	Bagmati, Ganga	14	63	959	10.890	0.100	0.1520	0.0297	9218	9419	36	31
2005	Bagmati, Gandak	12	81	1464	21.040	5.350	0.3343	0.1261	1031	4507	58	4
2004	Gandak, Bhutahi	20	211	9346	212.990	86.860	2.0990	0.6010	361017	568756	885	3272
2003	Gandak, Ganga	24	172	5077	76.020	11.960	0.9943	0.5140	43163	41261	251	108
2002	Bagmati, Kamala	25	6	8318	160.180	52.510	1.4446	0.5244	89021	329993	489	1450
2001	Gandak, Kosi,	22	194	6405	90.910	11.700	0.9042	0.2910	128419	93589	231	565
2000	Kosi	33	213	12351	90.180	8.090	0.6570	0.1476	183670	159421	336	2568

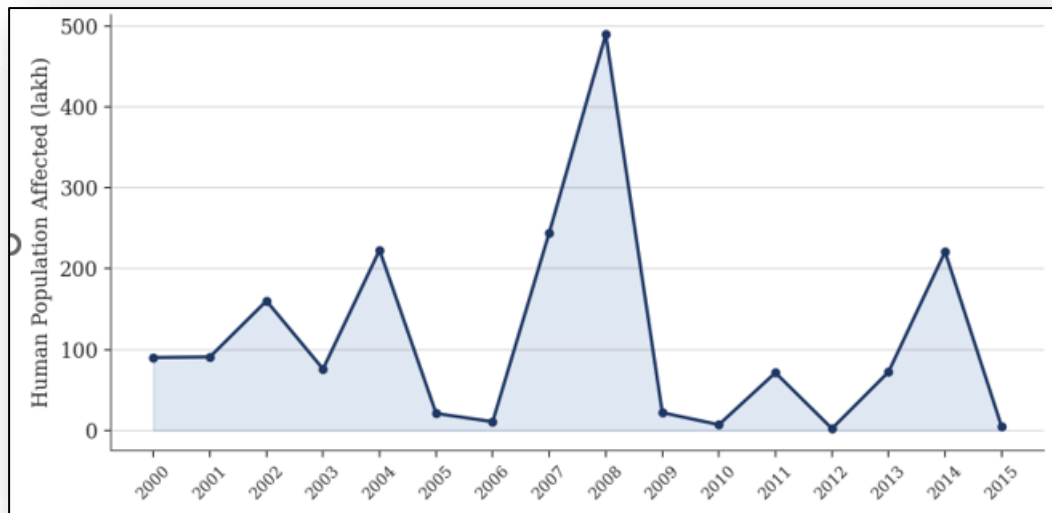


Figure 5. Flood-Affected Population in Bihar, 2000-2015. Author's visualisation of data presented in table 2 (Kansal, 2017).

Chapter 3: Institutional Framework for Flood Management

3.1 National Framework

India's national flood management framework is built around the broader disaster management system established under the Disaster Management Act, 2005, and operationalized through the National Disaster Management Authority (NDMA), the National Disaster Management Plan (NDMP), and supporting technical institutions such as the Central Water Commission (CWC) and the India Meteorological Department (IMD). Within this framework, flood management is treated as a combination of mitigation, preparedness, response, and recovery, rather than as a purely emergency function ([NDMA, 2008](#))

These guidelines reveal the enormity of the problem, stating that out of the total land area of India, which is 329 million hectares, 40 million hectares are flood-affected. The average area of the land that goes under water during floods is 75 lakh hectares annually. According to NDMA, flood-related damage has increased significantly in recent years. The average annual loss caused by flood between the year 1953 and 2005 was ₹1805 crores, while it increased between 1996 and 2005 to ₹4745 crores. ([NDMA, 2008](#))

3.2 State-Level Institutions in Bihar

At the state level, flood management in Bihar is coordinated through a multi-institutional framework led by the **Bihar State Disaster Management Authority (BSDMA)** and the **Disaster Management Department**. BSDMA is the apex statutory body for disaster governance in the state and was constituted under the Disaster Management Act, 2005. Its main responsibilities include laying down the State Disaster Management Policy, approving the State Disaster Management Plan, approving departmental disaster management plans, issuing guidelines for mainstreaming disaster risk reduction, recommending funds for mitigation and preparedness, and coordinating implementation of the state plan ([NIDM, 2022](#)).

The **Disaster Management Department** functions as the nodal department of the Government of Bihar for both natural and human-induced disasters. It is responsible for prevention, mitigation, response, relief, rehabilitation, reconstruction, legislation, policy-making, and capacity building. In the scenario of floods, the office has an important job in the implementing of preparations for floods and in organizing help, as well as in ensuring a district-oriented approach to responding to floods ([DMD, 2022](#)).

The **Water Resources Department (WRD)** of Bihar has the main function of being the technical body in charge of floods. In its use, the department fulfills the construction and maintenance of irrigation systems and flood prevention infrastructure, and regulates wastewater treatment in several municipalities of the state. In terms of floods that are common in Bihar, the activities of the Water Resources Department are critically connected with monitoring the behavior of rivers ([WRD Bihar, 2023](#)).

Within the WRD, the **Flood Management Information System / Flood Management Improvement Support Centre (FMISC)** aids in flood forecasting, sediment management, flood prevention, and making informed decisions. FMISC is especially relevant in Bihar as floods in this region are a consequence of river sedimentation, embankment stress, and trans-boundary flow conditions ([NWDP, 2025](#)). This institution is intended to bridge the gap between engineering measures and operational flood management decisions.

Capacity building and technical training are further supported by the **Bihar State Disaster Management Authority (BSDMA)**. Therefore, BSDMA has been established to facilitate disaster

risk reduction through training, research, and institutional support for government bodies, institutions, and communities. Its significance lies in the fact that in the case of flood management, not only infrastructure but also qualified specialists, trained rescuers, and well-informed local governance play a huge role in this process. ([NIDM, 2022](#))

Overall, Bihar's state-level flood governance system is structured around policy coordination, technical river management, capacity building, and district-level implementation. While BSDMA provides the strategic framework, the Disaster Management Department and WRD handle execution, FMISC supports technical forecasting and control, and DDMA operationalize flood response at the district level. This layered institutional arrangement is essential for a state like Bihar, where flooding is recurrent, spatially uneven, and closely tied to both river dynamics and governance capacity ([SDMP, 2025](#)).

3.3 District Administration and Local Governance

Bihar's district administrative structure is organised into **8053 Gram Panchayats, 533 Panchayat Samitis and 38 Zila Parishads, which are functional in the State, creating** a layered system for implementation and supervision ([Panchayati Raj Department, Government of Bihar](#)). This structure is especially important in flood-prone regions because district administrations must respond quickly to local variation in river breach, waterlogging, embankment failure, and evacuation needs.

At the district level, **District Disaster Management Authorities (DDMAs)** are responsible for preparing and monitoring District Disaster Management Plans and coordinating implementation of national and state disaster policies. They form the operational link between state institutions and local administration, ensuring that evacuation, relief distribution, shelter management, and recovery activities are implemented on the ground. This district-level structure is especially important in Bihar, where flood impacts vary across districts and response must be geographically targeted. ([BSDMA, 2025](#))

Local governance in Bihar is also shaped by the **Panchayati Raj system**, which provides a three-tier rural structure consisting of **Gram Panchayat, Panchayat Samiti, and Zila Parishad**. This system is vital in flood management because gram panchayats are often the first local institutions to report flooding, mobilize residents, identify vulnerable households, and support evacuation or relief distribution ([CAG, 2017](#)).

3.4 Role of SDRF, NDRF, and Line Departments

The **NDRF** plays a critical role during Bihar floods by deploying trained rescue teams to affected districts, carrying out evacuations, rescuing stranded people, and saving livestock. For example, during flood situations in Bihar, **25 NDRF** teams were deployed across 17 districts, where they rescued **67** persons, evacuated **2,691** people to safety, and saved 47 livestock ([NDRF, 2025](#)). This shows that the NDRF is not only a technical response force but also a life-saving institution essential to large-scale flood operations in Bihar.

The Bihar State Government has instituted a battalion of the **Bihar State Disaster Response Force (SDRF)** on the pattern of NDRF. Apart from the staff, suitable equipment and materials would also be provided to it to enable it to render help and extension services when disaster strikes it. It would train its human resource in disaster response based on the disaster involved and teach it how to act promptly on learning about the disaster ([BSDMA, 2016](#)).

Line departments are equally important because flood management is not limited to rescue alone. Departments such **PHED, Health, Education, Food and Consumer Protection, RDD, UDD, Social Welfare, and DMD**. ([BSDMA, 2021](#))

- **Disaster Management Department (DMD):-** Acts as the nodal agency for overall coordination, policy-making, and disaster response ([DMD, 2020](#))

- **Food and Consumer Protection Department:** - Coordinates the immediate distribution of food grains, dry rations, and subsidized commodities to affected families. ([BSDMA, 2021](#))
- **Rural Development Department (RDD):-** Enhance the wages for **Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)** for the repairs and restoration works undertaken as part of disaster recovery by Rural Development Department. ([BSDMA, 2021](#))
- **Social Welfare Department:** - Focuses on vulnerable groups including children, pregnant women, lactating mothers, and the elderly ([BSDMA, 2021](#)).
- **Health Department:** - Mobile Medical Team and Medical Camps need to be established at major flood relief and shelter locations. Furthermore, special medical teams must be deployed in mobile format. Each mobile team must be linked with two or three flood relief and shelter locations. The medical staff should be positioned at the designated location before the scheduled time ([Mega abhyan, 2024](#)).

3.5 Role of Community Institutions and NGOs

Community institutions and non-governmental organizations (NGOs) play a crucial role in flood management in Bihar because they operate at the household and village levels, particularly where access, communication or response capacity of government is constrained during floods. In flood-prone districts, community-based disaster risk reduction programmes have helped establish village disaster management committees, school disaster management committees, women's self-help groups, and local volunteer networks that improve awareness, preparedness, evacuation planning, and recovery support ([UNICEF, 2017](#)). Villagers were trained to keep ready flood response kits by the committee. Flood response kits comprise dry rations, necessary documents, and first-aid kits. The significance of this training lies in ensuring that families evacuate without wasting time by taking the essential items with them ([OXFAM, 2018](#)).

Chapter 4: Flood Preparedness and Response Systems

4.1 Early Warning Systems

Early warning systems are a core part of flood management in Bihar because they provide advance information on rising river levels, heavy rainfall, and likely inundation. In Bihar, the Flood Management Improvement Support Centre (**FMISC**), the Central Water Commission (**CWC**), the India Meteorological Department (**IMD**), the district administration, and community stakeholders all contribute to the warning chain. These systems are designed to improve lead time so that vulnerable households can evacuate and protect life and assets before floodwaters arrive ([Vyas J, 2021](#)).

4.2 Preparedness Measures

A key component of preparedness is the **early warning system**. **FMISC** is gathering live data via its **GIS** (Geo Information System) tools to obtain high-resolution images from satellites. **When** there is a noticeable rise in the water level, the district authorities in the downstream have been notified, which helps the residents of flood-affected low-lying areas move to safer areas ([UNDRR, 2021](#)).

The daily water levels measured at various river locations in the country are communicated by **Central Water Commission** at 6.00 AM with a one-day forecast. The forecasts are based on the correlation of the water level data through different gauges. This has been proved to be about 90% accurate ([Vyas J, 2021](#)).

4.3 Evacuation and Shelter Management

Evacuation management in Bihar means moving people from flood-prone areas to safer places before water levels become dangerous. It is usually carried out by the district administration with help from SDRF and NDRF teams, especially in villages that are slow to leave ([NDRF, 2019](#)).

During major floods, authorities use boats, relief camps, and local warnings to shift families and protect lives. In one Bihar flood response, 46,500 people were evacuated, showing that rapid evacuation is a key part of flood disaster management in the state ([Hindustan Times, 2017](#)).

Shelter management in Bihar typically involves the creation of **relief camps, temporary shelters, and community kitchens**. During the 2020 floods, the Bihar government reported that relief shelters were erected and 271 community kitchens were operating to support around 1.2 lakh people, while NDRF and SDRF assisted with evacuation ([Times of India, 2020](#)). In the 2019 Bihar urban floods, relief measures included temporary accommodation, safe drinking water, dry food, and sanitation support in flood-affected slums, showing that shelter management must address both physical safety and basic survival needs ([Sphere India, 2019](#)).



Figure 6. NDRF personnel engaged in evacuating marooned people in Araria. ([Hindustan Times, 2017](#))

4.4 Relief Distribution Mechanisms

Relief distribution in Bihar floods is mainly done through **direct cash transfer, relief camps, community kitchens, and distribution of essential items**. The state government transfers gratuitous relief directly to beneficiaries' bank accounts through PFMS/DBT, which improves speed and transparency in delivery ([Times of India, 2019](#)). In addition to cash support, district administrations and partner agencies distribute dry ration, polythene sheets, hygiene kits, medicines, and shelter materials to affected families ([Sphere india, 2024](#)).

UNICEF's flood response in Bihar also began with pre-positioned relief items such as tarpaulins, ORS sachets, halozone tablets, toilet kits, and hygiene kits for quick emergency support ([UNICEF, 2008](#)).

4.5 Inter-agency Coordination During Response

Meetings of the **Technical Advisory Committee (TAC)** and the **State Review Committee (SRC)** happen under the supervision of the **Water Resource Department (WRD)**. Materials for flood fighting are being stocked at critical locations near the rivers, i.e. sand bags, geo bags and nylon crates. Noticeable facilities for real-time communication are readily available through the VHF wireless equipment ([Vyas J., 2021](#)).

Chapter 5: Supply Chain and Logistics Assessment

5.1 Relief Supply Chain Architecture

The PDS in Bihar demonstrates how supply chain architecture directly impacts relief distribution. [Ray and Chaudhary 2026](#) identify procurement, storage, transport and distribution-related constraints that may affect supply-chain performance. Their direct applicability to emergency flood-relief operations requires separate assessment. While integrating technology (e-POS and digital verification) improves transparency and reduces theft, physical infrastructure (warehousing and fleet management) remains critical. A resilient relief supply chain architecture requires combining digital tracking with infrastructure upgrades to ensure timely, leak-proof delivery to beneficiaries.

The researchers propose an analytical chain linking procurement timeliness, storage integrity, transport reliability, distribution consistency and beneficiary satisfaction. The strength of these relationships remains to be tested using larger samples and formal statistical analysis ([Ray and Chaudhary 2026](#)).

5.2 Procurement and Warehousing Practices

For flood relief procurement in Bihar, the state uses **advance procurement and emergency procurement** supported by Bihar's public procurement rules, which allow emergency procurement for disaster management and permit fast-track purchase methods when urgent relief is needed ([Uncomplicate, 2024](#)).

The flood relief operations in Bihar involves the use of district-specific warehouses that store supplies for relief operations such as tarpaulin, tents and other emergency supplies. Some of the good practices in warehousing include storing the supplies at a height away from water, keeping them dry, recording stocks in the stock book and disposing off old or damaged supplies ([Live Hindustan, 2026](#)).

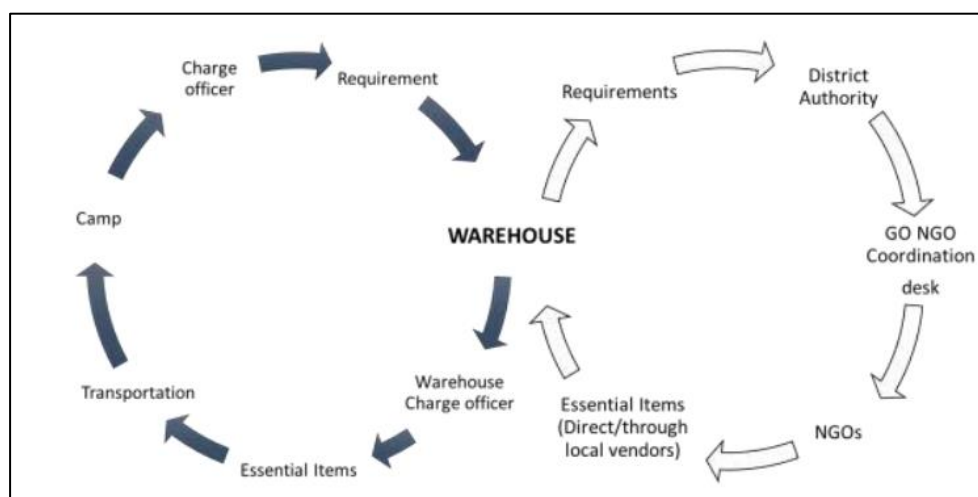


Figure 7. Process of Relief Distribution ([Sphere India, 2024](#))

5.3 Transportation Challenges During Flood Events

Floods in Bihar make transportation extremely inconvenient. Roads, rail lines, and other important routes often remain submerged or damaged by floods. Even for major highways and inter-district connections, floods can cut off routes from one point to another ([Times of India, 2020](#)).

This interruption causes logistics problems for humanitarian organizations, which may require the use of boats or tractors for getting to the affected villages. In certain instances, the respectable amount of time is required for delivering humanitarian aid due to unpassable roads. ([Médecins Sans Frontières, 2020](#)).

5.4 Last-Mile Delivery Issues

Accessing the impacted areas with humanitarian aid becomes challenging in Bihar floods as roads go out of service and the affected areas remain isolated. Relief teams have to use various means of transport in order to deliver food and medications to the marooned communities. ([Médecins Sans Frontières, 2025](#)).

5.5 Potential Areas for Optimization

Bihar can improve flood relief logistics by optimising warehouse locations, transport routes, and pre-positioned stock near high-risk districts to reduce response time and delivery cost ([Lopane et al., 2019](#)).

Chapter 6: Technology Applications in Flood Management

6.1 Use of GIS and Remote Sensing

GIS and remote sensing have gained immense popularity in Bihar due to their use in mapping flooded regions, monitoring flooding, and assisting with relief planning. The combination of satellite data and GIS technology provides the capability to identify at-risk districts, assess flooded farms, and produce flood hazard maps ([Manjusree et al., 2015](#)).

6.2 Flood Forecasting Tools

The flood forecasting techniques practiced in India encompass hydrologic, hydrodynamic models, satellite monitoring, telemetry, and mobile forecasting systems. Recently, CWC has launched the near real-time flood warning services for five days in advance based on rainfall data provided by IMD, satellite rain estimates, and automatic rain gauge measurements every three hours ([PIB, 2022](#)).

The approaches have become necessary for Bihar, as they increase lead time, enabling district administrations to plan evacuations and relief. The CWC's FloodWatch India mobile application and the newly developed C-FLOOD system enhance real-time flood monitoring and warning dissemination through satellites and GIS technology ([PTI, 2025](#)).

6.3 Decision Support Systems

Decision Support Systems (DSS) allow flood management organizations to take advantage of forecasts. The DSS consists of interfaces for integrating forecasting data, GIS layers, and vulnerability data. ([Schneider, 2025](#)).

At present, modern DSS in India serve as alert systems for impact-based warning, resource allocation, and response coordination during floods ([PIB, 2025](#)).

6.4 Emerging Technologies and Innovations

Emerging technologies are making Bihar's flood management more predictive and data-driven. AI/ML models, satellite-based flood mapping, web-based inundation platforms, and IoT-enabled monitoring are increasingly used to improve forecasting, risk assessment, and response planning ([PIB, 2026](#)).

For Bihar, these innovations are especially useful because they can support faster warnings, better evacuation planning, and more precise relief targeting in flood-prone river basins. Modern tools such as C-FLOOD, AI-based flood forecasts, and decision-support dashboards show how technology is shifting flood management from reactive relief to proactive preparedness ([PIB, 2025](#)).

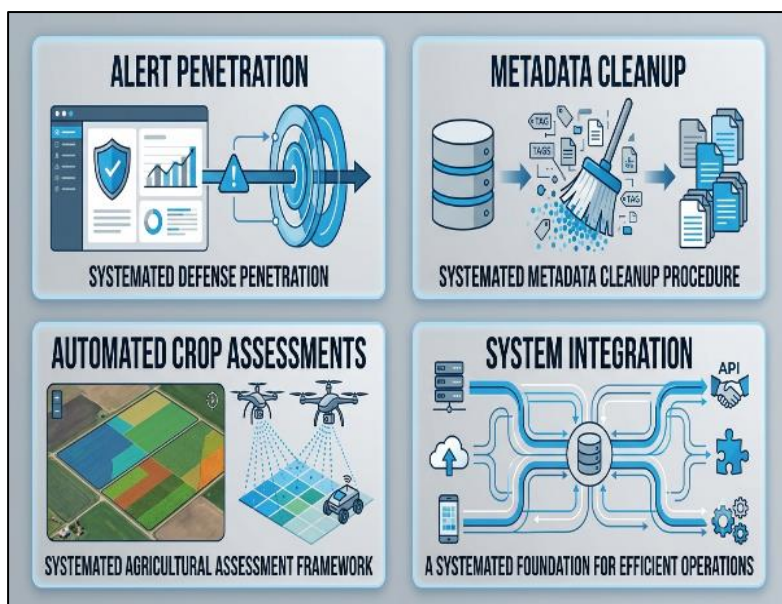


Figure 8. End-to-end tech workflow. (AI-generated)

6.5 Opportunities for Data-Driven Decision Making

There is a strong opportunity to improve Bihar’s flood management through **data-driven decision making**. Open-data platforms, risk score models, and flood datasets can help authorities match funding, planning, and relief more closely to actual flood losses and vulnerability across districts ([Open Contracting Partnership, 2025](#)).

In Bihar, this approach can help identify districts where flood impact is high but funding is low, improve allocation of SDRF resources, and support better forecasting and preparedness decisions ([Open Contracting Partnership, 2025](#)).

Chapter 7: Gap Analysis and Key Challenges

7.1 Institutional Challenges

The literature reviewed identifies Bihar's main institutional challenges as fragmented coordination, weak local capacity, poor community engagement, limited decision-making power at field level, and overreliance on structural measures like embankments. Some studies on flood governance in Bihar report coordination difficulties, gaps in the dissemination of warnings and constraints affecting local response capacity. ([Sinha et al., 2012](#))

The sources reviewed suggest that flood-related information is distributed across multiple agencies and that data-sharing and interoperability may constrain timely decision-making in some contexts. This weakens forecasting, preparedness, and coordination during floods, especially when rapid action is needed at district and community levels ([Sinha et al., 2012](#)).

7.2 Operational Challenges

Reported operational challenges in Bihar's flood management include delay in warning lead time, information flow, electricity and communication access, inter-agency coordination and last-mile dissemination.

Some field-based accounts suggest that communication between technical agencies, district authorities and communities can be uneven, affecting the final dissemination of alerts in particular locations. Studies from selected areas of North Bihar report instances in which warnings were received with limited lead time and thereby did not result in effective protective action. ([ICIMOD, 2026](#)).

7.3 Technology Gaps

Technology gaps in Bihar's flood management are mainly related to **short lead times, incomplete data integration, weak interoperability, and limited last-mile alert delivery**. Even though flood forecasting in India has improved, official sources note that exact prediction of peak flow and timing remains difficult, and the accuracy of flood advisories still depends heavily on rainfall forecast quality ([Teja & Umamahesh, 2020](#)).

7.4 Community-Level Challenges

Community-level challenges in Bihar include **low flood awareness, weak access to health and basic services, poverty, and limited local capacity to prepare for repeated floods**. Research from Darbhanga shows that rural flood-affected communities face continuing marginalization, food insecurity, disrupted health care, and barriers to services, which make local resilience difficult to sustain ([Kumar et al., 2016](#)).

A second challenge is **poor education and low participation in flood-risk reduction measures**. Studies in North Bihar note that lack of awareness, weak accountability, and poor understanding of flood management guidelines reduce the effectiveness of community action, even when government and NGO support is available ([Khan et al., 2023](#)).

7.5 Financing and Resource Constraints

In 2025, Bihar approved a Rs 4,415 crore World Bank-backed irrigation and flood management project, showing that major flood resilience initiatives still require external financing and multi-year investment ([Patna Press, 2025](#)). The approval of a major externally financed project indicates the importance of multi-year financing for irrigation and flood-management activities. Further

budgetary analysis is required to determine the extent of Bihar's dependence on external financing and the adequacy of available resources.

According to the cited media report ([Hindustan Times, 2026](#)), state authorities have sought additional relief packages, borrowing flexibility and central assistance for flood control and rehabilitation. While this may indicate a gap between needs and available resources, a thorough comparison of projected requirements, actual expenditure and programme outcomes is required to make recommendations for long-term prevention and infrastructure strengthening.

Table 3. Bihar State Budget - Key Fiscal Indicators. ([PRS, 2026](#)).

Items	2024-25 Actuals	2025-26 Budgeted	2025-26 Revised	% change (25-26 BE to 25-26 RE)	2026-27 Budgeted	% change (25-26 RE to 26-27 BE)	Annualised change from 24-25 to 26-27
Total Expenditure	2,81,939	3,16,895	4,23,284	34%	3,47,590	-18%	11%
(-) Repayment of debt	21,944	22,820	22,820	0%	22,665	-1%	2%
Net Expenditure (E)	2,59,996	2,94,075	4,00,465	36%	3,24,925	-19%	12%
Total Receipts	2,84,822	3,17,095	3,29,831	4%	3,47,753	5%	10%
(-) Borrowings	66,049	55,738	63,738	14%	61,939	-3%	-3%
<i>of which central capex loans*</i>	14,791	0	8,000	-	0	-100%	-100%
Net Receipts (R)	2,18,773	2,61,357	2,66,093	2%	2,85,813	7%	14%
Fiscal Deficit (E-R)	41,222	32,718	1,34,371	311%	39,112	-71%	-3%
<i>as % of GSDP</i>	4.2%	3.0%	11.8%	-	3.0%	-	-
Revenue Balance**	-357	8,831	-76,315	-964%	1,143	-101%	-
<i>as % of GSDP</i>	-0.04%	0.8%	-6.7%	-	0.1%	-	-
Primary Deficit	21,544	9,704	1,11,357	1047%	13,748	-88%	-20%
<i>as % of GSDP</i>	2.2%	0.9%	9.8%	-	1.1%	-	-
GSDP	9,91,997	10,97,264	11,39,595	4%	13,09,155	15%	15%

Chapter 8: Strategic Recommendations

8.1 Strengthening Institutional Coordination

Strengthening institutional coordination in Bihar requires clearer roles, better information sharing, and stronger links between the Water Resources Department, Disaster Management Department, district administration, and community-level actors. FMISC in Bihar was specifically created to improve coordination by providing flood information, emergency control-room support, connectivity, and technical links with stakeholder agencies, showing that institutional coordination depends heavily on shared data and common operational platforms ([FMISC, 2012](#)).

Studies of India's flood governance also show that coordination fails when agencies work in silos, communication systems are incompatible, and responsibilities overlap. Improving coordination therefore means establishing unified command structures, regular joint drills, standardized protocols, and stronger inter-agency communication so that warnings, resources, and response decisions move quickly from the state to the district and village levels ([Sphere India, 2013](#)).

8.2 Improving Relief Supply Chains

Improving relief supply chains in Bihar requires **better transparency and needs-based allocation**. Past field reports describe uneven relief coverage and administrative or logistical delays in selected flood-affected areas. These findings support the need for transparent inventory systems, local monitoring and clearly defined distribution criteria ([Down To Earth, 2025](#)).

8.3 Technology Integration Roadmap

A practical technology integration roadmap for Bihar should move from **data collection to real-time warning, decision support, and community dissemination**. The central government's current flood and disaster systems already combine GIS-based DSS, telemetry, satellite data, AI/ML, and mobile alert platforms, so Bihar can build on these tools rather than start from scratch ([MOES, 2025](#)).

For Bihar, the roadmap should include: 1) expanding river and rainfall monitoring stations, 2) integrating FMIS, CWC, IMD, and NRSC datasets into one dashboard, 3) using GIS-based inundation forecasts for district planning, and 4) sending geo-targeted alerts through SACHET, FloodWatch India, and local communication channels ([PIB, 2026](#); [NDMA, 2026](#)).

8.4 Public-Private Partnerships

Public-private partnerships can strengthen Bihar's flood management by bringing private expertise, finance, and technology into areas where the state's capacity is limited. PPP models are especially useful for resilient infrastructure, because they help share risk, support financing, and improve procurement, monitoring, and maintenance in disaster-prone settings ([World Bank, 2019](#); [Eyerkauffer, 2016](#)).

For Bihar, this means PPPs can be used for smart drainage, flood-warning systems, relief logistics, and risk-aware infrastructure, as long as roles, responsibilities, and disaster-risk allocation are clearly defined. Recent research also shows that blended PPP models with viability gap funding and performance-based contracts can improve climate resilience and make flood-control projects more financially viable ([Madapala and Ganesh, 2025](#); [Rao, 2026](#)).

Table 4. Summary of Financing Mechanisms ([Madapala and Ganesh, 2025](#)).

Financing Mechanism	Description	Pros	Cons
Blended Finance	Combines public and private funds to leverage resources	Reduces risk for private investors	Complexity in structuring deals
Resilience Bonds	Performance-based bonds for funding resilience initiatives	Incentivizes measurable outcomes	Requires robust monitoring frameworks
Public-Sector Guarantees	Guarantees to cover revenue or operational shortfalls	Attracts private investment	Government budget constraints
Risk-Pooling Mechanisms	Shared risk frameworks among stakeholders	Spreads risk, encourages collaboration	May dilute accountability

8.5 Capacity Building Measures

Capacity building measures in Bihar focus on **training officials, community volunteers, and local institutions** so they can respond quickly during floods.

Bihar has also strengthened community capacity by training “Yuva Aapda Mitras” and forming community emergency response teams, which improve local search-and-rescue and evacuation support. These initiatives show that capacity building in Bihar is not only about formal institutions but also about creating a trained grassroots response network ([TOI, 2026](#); [AMS, 2022](#)).

8.6 Community-Based Flood Resilience Approaches

Community-based flood resilience in Bihar relies on **local preparedness, community volunteers, flood shelters, grain banks, and early warning systems**. The Bihar DRR Roadmap recommends village-level Community Emergency Response Teams, flood shelters at Gram Panchayat level, storage space for grains and fodder, and no-construction zones based on flood vulnerability mapping ([UNICEF, 2015](#)).

Chapter 9: Conclusion

The review indicates that Bihar has developed important institutional and operational capacities for flood response, including the involvement of the Water Resources Department, Disaster Management Department, FMISC, NDRF, SDRF and local administrative bodies. At the same time, the sources reviewed identify continuing concerns regarding coordination, data sharing, warning dissemination, relief logistics and the consistency of local implementation. These concerns should be understood as evidence of uneven capacity across locations and situations rather than as proof of uniform institutional failure. Further field-based research, including district-level comparisons and interviews with officials and affected communities, would be required to determine the scale and causes of these limitations.

9.1 Limitations of the study

- Based entirely on secondary data; no primary field survey.
- Findings are Bihar-specific and may not generalize without adaptation.
- Some data (e.g., SDRF allocation mapping) may not reflect the latest administrative changes.

9.2 Recommendations

Low-cost institutional fixes rather than new bureaucracy or engineering: shared data protocols/dashboards, decentralized stockpiling, and stronger community volunteer networks.

9.3 Future Scope

- AI/ML for predictive flood modelling and damage assessment
- GIS/digital-twin modelling of embankments and river behavior
- IoT-based river and rainfall sensor networks
- Satellite monitoring linked to village-level warnings
- Community-based early warning combining formal and local knowledge
- Blockchain for relief-tracking transparency (PFMS/DBT)
- Predictive analytics linking loss data to SDRF/budget decisions

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